Water Analyser



Category: **Laboratory Instruments & Glassware** Type: **Full SOP** Priority: **Medium** 1 labs

Covers these lab items: Water and juice analysis machine

What it is used for

A water analyser measures selected water-quality parameters, and some models also analyse juice or other liquid samples. Use the correct test method, sensor, and calibration standards for each parameter to obtain reliable results.

Main parts

- Water analyser main unit and display
- Parameter-specific sensors, electrodes, or probes
- Sample cells, cuvettes, or test vessels
- Reagents and test kits where required
- Calibration standards and reference solutions
- Sample holder and temperature sensor, if fitted
- Power supply, batteries, and data recording system

Before you start

1. Identify the sample type and the parameters to be tested.
2. Read the instrument manual to confirm the approved method and sample compatibility.
3. Inspect the instrument, sensors, cables, and sample cells for damage or contamination.
4. Prepare fresh reagents and calibration standards according to the instructions.
5. Calibrate each required parameter using the specified standards.
6. Collect a representative sample in a clean, suitable container.
7. Bring samples and standards to the required temperature if the method specifies it.

How to use

1. Switch on the analyser and select the required test parameter or method.
2. Rinse reusable probes or sample cells as specified by the method.
3. Prepare the sample and add reagents when required, following the test-kit instructions.
4. Insert the sample cell or immerse the probe to the specified depth.
5. Wait for the required reaction time or sensor stabilisation period.
6. Read the result and confirm that it falls within the instrument's measurement range.
7. Record the sample identification, parameter, result, units, temperature, and test method.
8. Repeat the test or use a suitable dilution when the result is outside the valid range.
9. Dispose of used samples and reagents according to laboratory procedures.
10. Clean the probes and sample cells, then switch off and store the instrument correctly.

⚠ Safety

- Wear laboratory coat, suitable gloves, and eye protection when handling samples or reagents.
- Never taste test samples, including juice or water of unknown quality.
- Follow reagent labels and safety data sheets for handling and disposal.
- Keep liquids away from electrical connections and the instrument body.
- Do not mix reagents unless the approved test method instructs you to do so.
- Treat potentially contaminated water samples as hazardous and wash hands after testing.

Care & storage

- Rinse and dry sensors or sample cells as recommended for each parameter.
- Store electrodes and probes in the specified storage solution when required.
- Keep reagents within their expiry dates and follow storage instructions.
- Check calibration and perform routine quality-control checks using reference standards.

Common mistakes to avoid

- Using expired reagents or incorrect calibration standards.
- Testing a sample with the wrong method or incompatible sensor.
- Contaminating samples through dirty cells, probes, or containers.
- Ignoring reaction time, temperature requirements, or instrument measurement limits.
- Recording results without units, sample identification, or calibration checks.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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